Jungsoo Kim

Ph.D. student, Graduate School of Data Science, Seoul National University

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Education

Master — **Seoul National University, Graduate School of Data Science**, Data Science — Seoul, Korea — Mar 2022 – Aug 2024
- GPA: 4.22/4.5
- Courses: Convex Optimization, High Performance Computing, Big Data and Knowledge Management System, Causal Inference
- Teaching Assistant: Causal Inference, Machine Learning and Deep Learning II, Basic Statistics & Mathematics Bootcamp

B.P.A. / B.B. — **Yonsei University, Underwood International College**, Science, Technology and Policy; Quantitative Risk Management — Seoul, Korea — Mar 2017 – Feb 2022
- GPA: 4.15/4.5
- Courses: Mathematical Statistics, Regression Analysis, Econometrics, Survival Data Analysis, Time Series Analysis, IT and Socio-Economic Change, Principles of Financial Engineering

Exchange Student — **National University of Singapore, Yale-NUS College**, Mathematical and Computational Science — Singapore — Sept 2019 – Feb 2020

Publications

Decomposing Conformal Uncertainty: Calibration- and Instance-Driven Feature Attribution — Jan 2026
Accepted as a poster.
(Advances in Neural Information Processing Systems (NeurIPS))

A Tale of Two Uncertainties: Global-Local Attribution for Conformal Prediction — Jan 2026
Spotlight paper.
Sangyeon Cho, Minyoung Cho, *Jungsoo Kim*, Sujeong Oh, Sanghack Lee
openreview.net/forum?id=2alvedJMpp (ICML 2026 Workshop on Epistemic Intelligence in Machine Learning (EIML))

Instrumental Variable Representation Learning under Confounded Covariates — Jan 2025
Jungsoo Kim, Kwonho Kim, Inwoo Hwang, Sanghack Lee
(NeurIPS 2025 Workshop on CauScien: Uncovering Causality in Science)

Diagonal Argument: Anthology of 5th POSTECH SF Award-Winning Works — Jan 2025
Chapter: *Will of Stochastic Ghost*. ISBN: 9791167375810.
Geonryul Park, Huyeong Lee, *Jungsoo Kim*
product.kyobobook.co.kr/detail/S000217575854 (Eunhengnamu)

Research and Industrial Experience

Ministry of Food and Drug Safety (MFDS) Project, Project Lead for Causal Inference Section — Mar 2025 – Dec 2025
- Led an MFDS project on drug-effect estimation using conformal prediction.
- Developed methods for interpretable uncertainty quantification as part of a research team.

SK Center for Social Value Enhancement Studies, Research Assistant — Aug 2021 – Jan 2022

- Reviewed beneficiaries' financial records totaling 10 billion KRW and estimated the next fiscal year's budget.
- Researched trends in the Korean social economy, focusing on economic polarization and building a research database.

Mooncademy (Korean Traditional Music Education Start-up), Product Manager — Oct 2020 – Apr 2021

- Planned website information architecture and page layouts for a Korean traditional music platform, collaborating with engineers and designers.
- Launched a Wadiz crowdfunding campaign for gayageum lectures, raising KRW 10,491,700.
- Presented the 2021 annual plan at an investor relations meeting.

Leadership and Involvement

Coding Instructor for Malaysian High School Students — 2025-07 to 2025-08
Taught Python programming and basic algorithms through an SNU Language Education Institute outreach program. Designed interactive sessions emphasizing teamwork and oral presentation to discourage reliance on generative AI tools and foster authentic problem-solving.

Manager and Main Speaker for STP-STS Forum — May 2022
Organized and hosted a department forum for students and faculty in Science & Technology Studies at Seoul National University and Science & Technology Policy at Yonsei University, while researching legal controversies in medical law regarding tattooing.

Project Management Director at Social Innovation Creators' Academy — 2019-03 to 2020-06
Managed cross-team project schedules and deliverables, coordinated milestones across multiple teams, and developed a semester-long project roadmap.

Yonsei Business College Drama Club (Yonsang-geuk-u-hoi) — 2017-03 to 2021-10
Led production planning and mediated creative discussions across five productions. Designed a fully remote event in response to COVID-19 constraints.

Vice President in Major Student Council — 2017-11 to 2018-11
Liaised with faculty on curricular and administrative issues and planned major information sessions and inter-major forums on science fiction.

Honors and Awards

5th POSTECH SF Award, Excellence Prize — Jan 2025
Authored a short story exploring conflicts surrounding generative-AI-based evidence.

Journal of Korea Artificial Intelligence Association, Best Paper — Jan 2024
Submitted a master's project paper on instrumental variables.

CSES Social Value Research Idea Award, Excellence Prize — Jan 2024
Proposed a study on causal structure analysis for Korean population policy.

LH COMPAS Land & City Data Competition, Excellence Prize — Jan 2024
Identified medically underserved areas in Asan through geospatial analysis.

EdTech Data Analysis Hackathon, Grand Prize — Jan 2023
Built a dropout-prediction model using survival analysis.

Public BigData Analysis Hackathon, Excellence Prize — Jan 2020
Developed a traffic-accident risk model with XGBoost and interpreted drivers using SHAP.

Art Data Hackathon Jan 2020
Designed a traditional musical-instrument recommendation system.

Skills and Language

Programming and Data Systems: R, Python, C++, C, SQL, Neo4J, BigQuery, Kafka

Machine Learning and Causal Inference: Fine-tuning super-resolution models, training Stable Diffusion with PyTorch, geospatial data analysis with GeoPandas and QGIS, instrumental variable analysis, conformal prediction

Languages: Korean, English (TOEFL 100, OPIc AL)

Productivity: Microsoft Word, PowerPoint, Excel, Overleaf